On confidence interval composition in short-term forecasting of electricity consumption

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***Abstract*—The featured article describes the problem of confidence interval calculation of the forecast electricity loads field. The methods of calculation of confidence interval for forecasting in electricity loads field introduced.**

Keywords—electricity loading, forecast electricity loads, confidences intervals.

I. INTRODUCTION

The process of forecasting power consumption is an important task for the subjects of the wholesale electricity and power market (hereinafter referred to as the WECM subject). Works [1] – [6] have already been devoted to this problem. The result of solving such problems is the forecast of power consumption (hereinafter referred to as the ES), extended by L - values ahead, where ($L \geqslant 1$ hour). Denote the process of power consumption by the function $V(t)$, where t is the time interval, then $V(t + L)$ is the time series extended by L values ahead. Usually, in practice, the extension of the time interval on the interval $L = 24$ is required. As a rule, the subject of the WECM has an established method for implementing a short-term forecast of power consumption [7], [8].

At the same time, the expert makes a decision on the correctness of the data obtained, assessing the risks of using one or another method of short-term forecasting of ES. To make such a decision, the expert needs to have additional characteristics that describe the properties of the obtained forecast. One of these characteristics is the geometric mean deviation value for assessing the accuracy of the predictive model. However, this is also not enough, since the characteristic shows a common time series error.

Another, more appropriate characteristic can be a relative measure η – the number of cases where the actual implementation was covered by the interval forecast of ES with respect to the total number of ES forecasts,

$$\eta = \frac{\rho}{\rho + q} \tag{1}$$

where ρ – is the number of forecasts confirmed by actual data and is the number of forecasts not confirmed by actual data. This characteristic is often used when the forecast data of the previous period were not documented.

This characteristic is often used when the forecast data of the previous period were not documented. To use this characteristic, it is necessary to build confidence intervals of the forecast. Moreover, the confidence interval displayed graphically in L steps gives the expert an opportunity to evaluate not only the value of the forecast, but also his confidence interval with a predetermined degree of accuracy. It is known from [9] that there are several ways to construct such an interval. Confidence intervals differ in the type of envelope function. Consider these functions.

In general, the confidence interval is determined by the formula [10]:

$$s_y K^* = t_\alpha s_y \sqrt{1 + \frac{1}{n} + \frac{1}{\sum t^2} t_L^2} \tag{2}$$

where n – the number of measurements on which the forecast is based, L – number of samples for which the trend is extended, t_α – the value of Student's t-statistic, s_y – trend standard deviation, t_L – time extended by L trend, $t_L = n + L$ and t ranging from 1 to n.

If we substitute $t_L = n + L$ in this formula, we get that the value K^* depends on the lead time, as the observation duration n increases, the values K^* decrease. As the L value increases, the values increase. This formula describes a linear confidence interval.

II. EXPERIMENT

Consider ways to build confidence intervals, depending on the type of the envelope function. Below are the functions of building confidence intervals based on the polynomial model (3) and the hyperbolic cube (4)

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$$s_y K^* = t_\alpha s_y \sqrt{1 + \frac{1}{\sum t^2} t_L^2 + \frac{\sum t^4 - \left(2\sum t^2\right) t_L^2 + n t_L^4}{n\sum t^4 - \left(\sum t^2\right)^2}} \tag{3}$$

$$s_y K^* = t_\alpha s_y \sqrt{1 + \frac{1}{\sum t^2} t_L^2 + \frac{\sum t^4 - \left(2\sum t^2\right) t_L^2 + n t_L^4}{n\sum t^4 - \left(\sum t^2\right)^2} + \frac{(\sum t^6 - 2\sum t^4) t_L^2 + (\sum t^2) t_L^6}{\sum t^2 \sum t^6 - (\sum t^4)^2}} \tag{4}$$

Fig. 1 shows the dependence of the coefficient K^* with an increase in the forecast interval K^* $(t + L)$, where L is in the range from 1 to 30 hours.

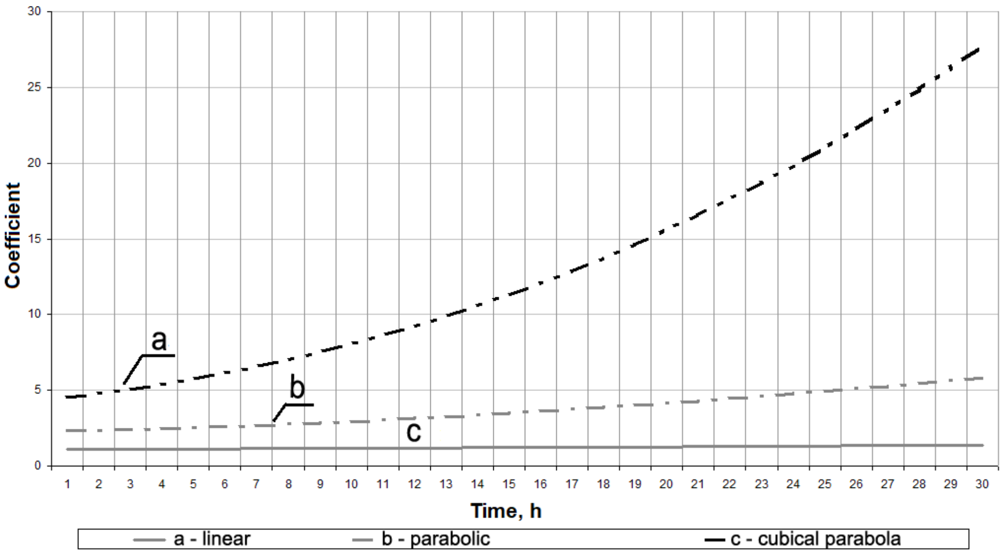

Fig. 1. The dependence of the coefficient K^* on the duration of the forecast.

To build confidence intervals for short-term forecasting of ES, it is necessary to extend the trend line of the time series of ES and, simultaneously with the extension of the trend line, build confidence intervals calculated on the basis of the deviation of the actual data from the forecast. As a method for predicting an ES, we take the moving average method [11] as a basis and extend the trend line by 10 counts (1 count - 1 hour). By calculating the standard deviation (hereinafter referred to as the standard deviation) on the basis of n = 20 samples, we construct confidence intervals using formula (1). The confidence intervals will be built 2: for a confidence probability p = 0.8 and for p = 0.9. According to the results of calculations, the graph was obtained in Fig. 2.

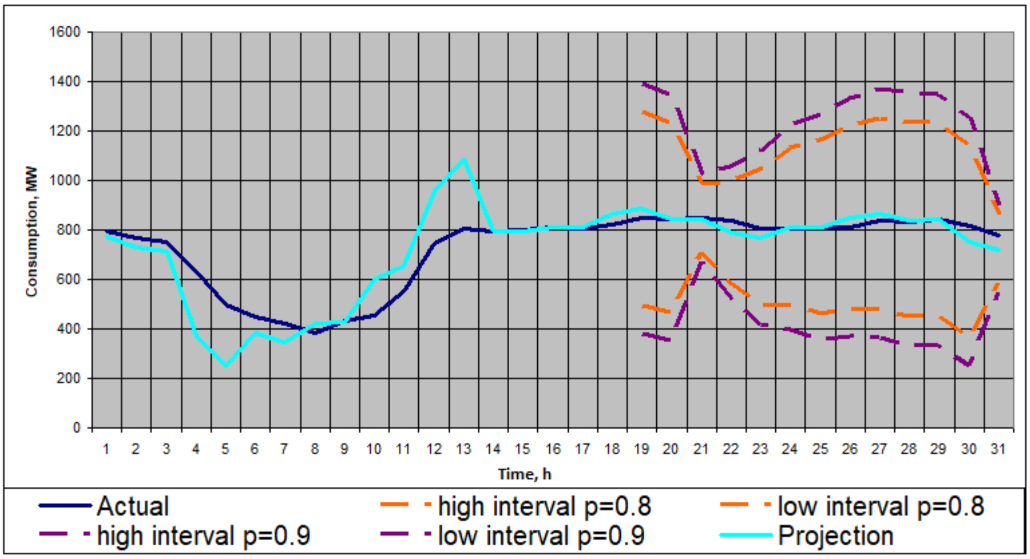

Fig. 2. Direct confidence interval when $n = 20$, $L = 10$.

From the analysis of the graph it follows: the moving average method is not suitable as a basic method of forecasting when constructing the confidence interval of the EA. The direct confidence interval is too narrow and when the trend reverses, the real value does not fall into it (area a on the chart). Thus, if a sharp increase in the trend is not expected, then for long-term forecasting it is necessary to combine this method with classical statistical methods.

In constructing the power consumption forecast in Fig. 3 – 4, the linear regression method was used [6]. Using this method, a forecast for the day was built, based on the consumption data of the previous three days. Fig. 3 – 4 show the forecast graphs in the interval of 160 hours. In this case, the graphs show the difference in the forms of representation of confidence intervals.

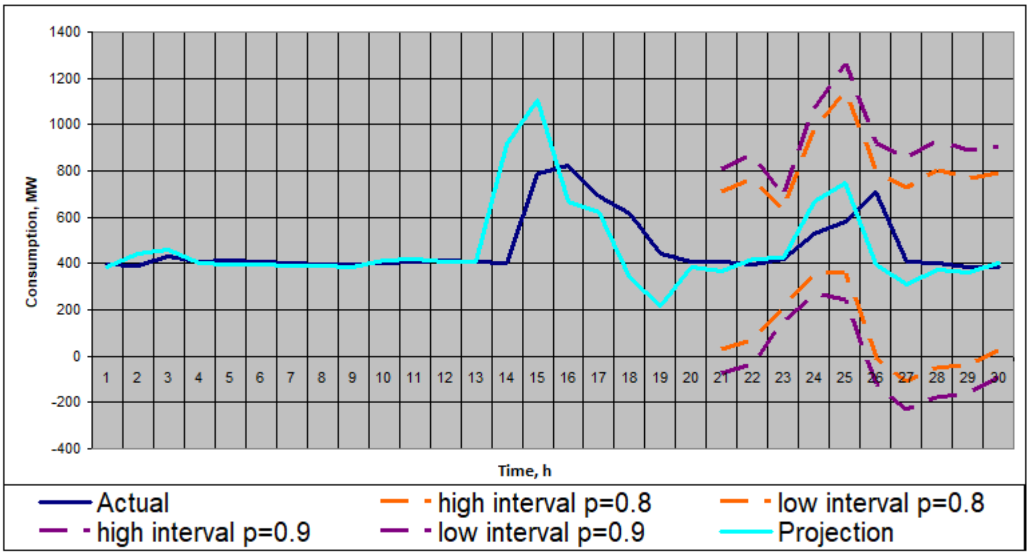

Fig. 3. Parabolic confidence interval when $n = 20$, $L = 10$.

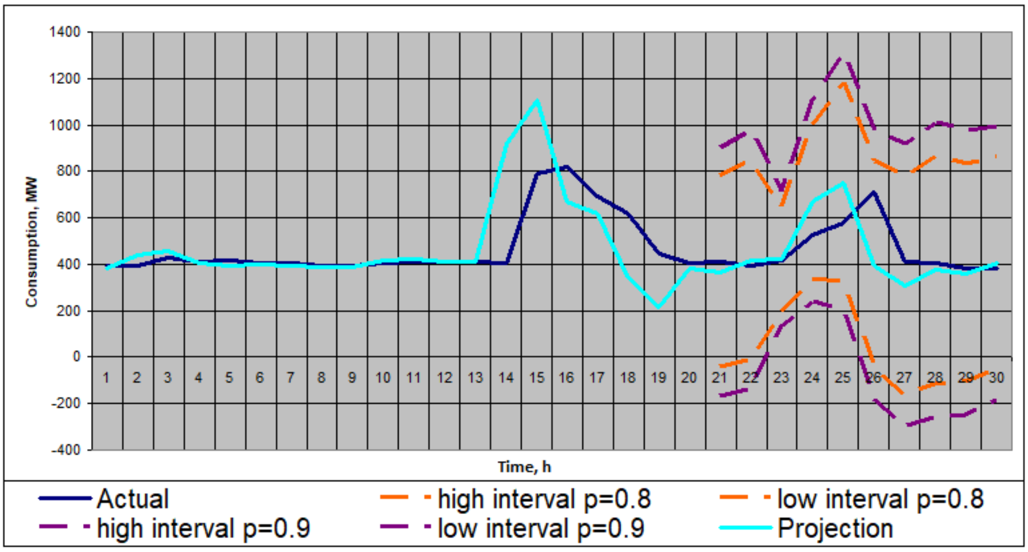

Fig. 4. Cubic parabola confidence interval when $n = 20$, $L = 10$.

As can be seen from the graph in Fig. 4, a method using a hyperbolic parabola is devoid of this disadvantage. All actual values fall within the limits of the confidence interval. However, it can be observed that even in the absence of differences, the confidence interval of a hyperbolic parabola is quite wide and it is advisable to use it only in certain cases, for example, with a sharp fall in trend.

Consider the use of different confidence intervals on short samples ($n = 20$) of data.

Confidence intervals for ($n = 20$, $L = 20$) are shown in Fig. 5. If the tangent trend line is shown to clearly show a decrease in the trend line, the method based on a second-degree parabola is more suitable for constructing a confidence interval, and abrupt trend change - a method based on a hyperbolic parabola.

For the graph in Fig. 5, the characteristic was calculated using formula 1. For the 2-day data, the following results were obtained. 3 values were not in the confidence interval ($q = 3$) and $p = 0.85$. When recalculated by the same method, data for the year in the confidence interval $q = 1337$ and $p = 0.846$.

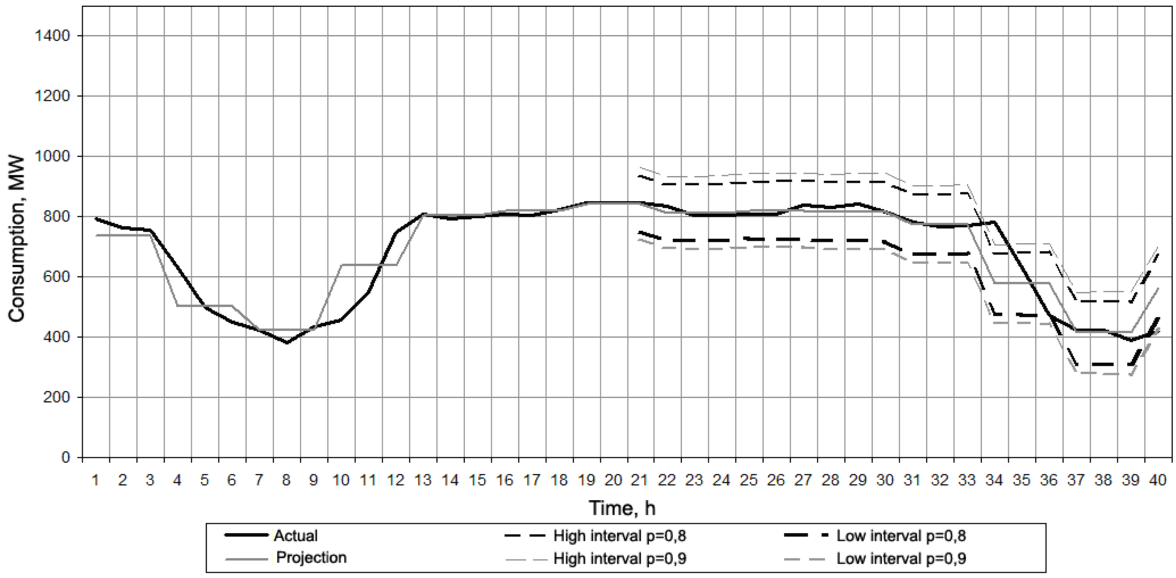

Fig. 5. Direct confidence interval when $n = 20$, $L = 20$.

The same measure of accuracy was obtained for the other forms of confidence intervals. This measure is equal to one if all real values fall into the confidence interval of the forecast, otherwise it is less than one. The results are presented in summary table 1.

TABLE I. THE NUMBER OF HITS IN THE CONFIDENCE INTERVALS

	Direct 2 days	Direct year	Parabolic 2 days	Parabolic year	Hyperbolic 2 days	Hyperbolic year
q	3	1337	1	650	0	72
p	0.85	0.845	0.95	0.925	1	0.99

If you analyze the data in the table, it is obvious that the method using the hyperbolic parabola function has the least overshoot of the confidence interval, but if you look at the graph in Fig. 6, you can see that the interval is just too wide, the upper limit of the interval allows a prediction error above 50% , and the use of such an interval cannot help the expert in the forecast.

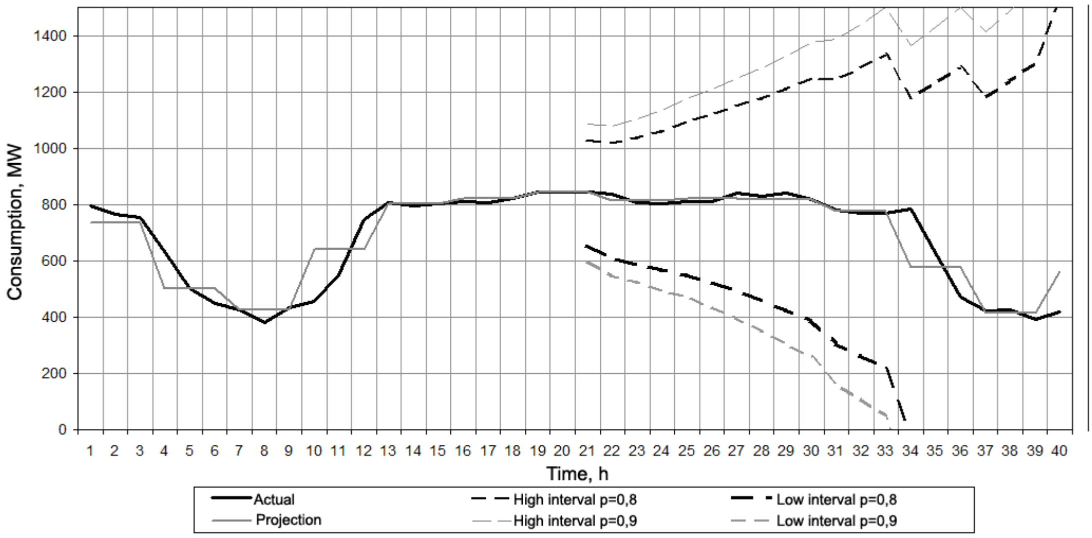

Fig. 6. Hyperbolic Parabola Confidence Interval when $n = 20$, $L = 20$.

III. CONCLUSION

Thus, it is advisable for an expert to present not only the expected values of the predicted value of the EA, but also the display of the graph of the obtained time series of the EV on the interval L, with the display of the confidence intervals. The question remains about the appropriate level of confidence level when constructing confidence intervals. Increasing the probability leads to a narrowing of the interval, so that actual values may more often go beyond its boundaries. If the probability level is too low, the interval, on the contrary, turns out to be excessively wide, which leads to an increase in uncertainty and makes it difficult to make decisions when planning modes of the electric power system. The optimal level of probability is selected based on the experience of using interval forecasts. The software implementation of the forecasting method should provide for the possibility of setting and changing the parameter of the confidence level. In addition, in cases where there is a significant increase / decrease in the EA, it is advisable to display on the chart several confidence intervals with varying degrees of probability.

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